of questions customers might ask, and which messaging channels—such as Messenger, Slack, voice-based assistants, or others—customers can use. Natural language processing capabilities built into the platform understand and learn the nuances and the context of conversations. And developers can use APIs to integrate the bot to back-end systems, as a way of pulling in data such as team schedules and seat availability.

Oracle's intelligent bots can support many of today's most popular interaction channels including messaging clients such as Facebook Messenger, Kik, Skype, Slack, and digital voice assistants such as Amazon Echo, Amazon Dot and GoogleHome. Additionally, Oracle's intelligent bots provide native and JavaScript SDKs to extend mobile and Web-based applications with chat and voice capabilities via Apple Siri, GoogleVoice or Microsoft Cortana.

### Q What is the scope of open source in chatbot development?

The core engine of a chatbot is powered by AI/ML components for intent recognition, natural language processing, etc. This is the domain of well-known open source projects such as TensorFlow, Stanford NLP, etc. Apart from these, open source projects such as Apache Spark and Kafka are also used by the bot engines.

### Q With chatbots gaining pace, what are the expected hiring trends?

Intelligent bots will transform every facet of every industry and dramatically improve the customer experience. In fact, chatbots can upskill employees with the most trending technology and also help them with relevant operations management. Apart from this, they can manage multiple tasks at the same time.

The skillsets needed for building chatbots are varied. These range from UX expertise (how do you engage a user in the absence of a GUI) to expertise on domains—to understand the user's psyche and the range of customer interactions.

### Q What technologies are Indian developers focusing on?

We see a very vibrant and active software development scenario in India, with developers working across the entire spectrum of technology. We find there's more developer appetite for cutting-edge technologies such as containers, chatbots, etc, when building innovative applications.

> *"This year, 'Oracle Code – Live for the Code' is scheduled to be held in Hyderabad and Bengaluru on April 4 and 10, respectively."*

There are three trends most noticeable among Indian developers. First, cloud native development using Microservices is set to go mainstream. Microservices based development is no longer a new buzzword, but an established best practice to deliver products quickly to the business in the digital era. This boosts agility, giving enterprises a significant competitive edge.

Second, real chatbot applications with natural language processing will become the norm this year. Using chatbots, existing applications can be extended to newer channels, while newer applications can interact in novel ways.

And third, Node.js and JavaScript will continue to build a massive following. JavaScript has become the ubiquitous language with both client and server-side programming capabilities. Most new applications are built using JavaScript frameworks such as React, Angular, etc, with Node.js on the backend. Polyglot applications continue to remain a focus for developers looking to choose from a multitude of programming languages.

Above all this, open standards, open platforms and open source remain a key priority, since developers want zero lock-in. In line with Oracle's commitment to open standards and open source, the company recently announced three new open source container utilities – Smith, a secure microcontainer builder; Crashcart, a microcontainer debugging tool; and Railcar, an alternative container runtime.

### Q How is Oracle engaging with the Indian developer community?

With the democratisation of coding, Oracle is committed towards developing, supporting, and promoting various technologies for developers. India is a critical market for us, and Oracle Cloud Platform validated for India Stack is a testimony to our commitment. We'll continue to engage with and empower the vibrant developer community here to build world-class products.

Oracle, which has been successfully running 'Oracle Code' globally, acts as a learning platform for technical experts and industry leaders, as well as a platform where developers get to exchange experiences. The company has brought its global flagship developer event to India, conducting it in Bengaluru and Delhi in 2017. Such events are an opportunity for developers to experience easy, modern and open cloud development technology with workshops and other live, interactive experiences and demos with global tech evangelists. Oracle Code not only acts as a learning platform but is used to exchange ideas and share experiences too.

To experience cloud development technology with workshops and other live, interactive experiences and demos, Oracle once again invites the developer community to participate in this event in 2018. 'Oracle Code – Live for the Code' is scheduled to be held in Hyderabad and Bengaluru on April 4 and 10, respectively. END